ChE 425 Nanotechnologies for Pharmaceutical Applications

Contract Hours: Lectures:

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Textbook and reading materials:
Reading materials from high-impact journals and a few textbooks, including but not limited to, Science, Nature, JACS, PRL, Biomacromolecules, ACS Nano, Nano Letters, Journal of Colloids and Interfacial Science…

Topical outline

<u>Background and Preparation</u>

Week 1-2. Factors that impact the developability of drug candidates.
- Challenges facing the pharmaceutical industry
- Factors that impact developability
- Remarks on developability
- Drug delivery factors that impact developability

Week 3. Evolution of controlled drug delivery systems.
- Biopharmaceutics and pharmacokinetics
- Material science
- Protein, peptide, and nucleic acids
- Discovery of new molecular targets
- Microelectronics and microfabrication technologies
- Nano-drug carriers

<u>Advanced Nanotechnologies and Microtechnologies for Drug Delivery</u>

Week 4-5. Self-assembled delivery vehicles: basic theoretical considerations and modeling concepts.
- Brief reminder of equilibrium thermodynamics
- Principles of self-assembly in dilute solutions
- Solubility and partitioning of drugs
- Ways to model interactions in colloidal systems
- Kinetics of drug transfer from mobile nanocarriers

Week 6. Nanoparticle characterization
- Dyanmic light scattering
- X-ray scattering
- Neutron scattering
- Microscopy

Week 7. Liposomes as intravenous solubilizers.
- Solubilizing vehicles with precipitation risk upon dilution
- Solubilizing vehicles maintaining solubilization capacity upon dilution
- Mechanistic release aspects

Week 8-9. Polymeric nanoparticle drug delivery systems.

- Safety and biocompatibility of polymers
- Micelles and micellization
- Encapsulation techniques
- Physicochemical properties of nanoparticles
- Drug loading
- Biodistribution and toxicity
- Targeting micellar nanocarriers: passive vs. active targeting

Week 10-11. Nanosuspensions with enhanced drug dissolution rates of poorly water-soluble drugs.

- Crystal growth and nucleation theory
- Creating supersaturation and stable nanosuspensions
- Antisolvent precipitation via mixer processing
- Antisolvent precipitation by using ultrasonication
- Nanoprecipitation using microfluidic reactors
- Particle engineering by spray
- Precipitation by rapid expansion from supercritical to aqueous solution

Week 12. Delivery of genes and oligonucleotides.

- Systemic delivery
- Cellular delivery
- Current and future approaches Week 13. Delivery of cells.
- Islet cell encapsulation and implant
- T-cell delivery

Pharmaceutical Fundamentals

Week 13. Physiological, Biochemical, and Chemical barriers to Oral Drug Delivery.

- Mechanistic PK-PD models
- Physiological barriers
- Biochemical barriers
- Chemical barriers
- Presystemic and first-pass metabolism

Week 14. Cell culture models for drug transport studies.

- Intestinal epithelium
- BBB
- Nasal and pulmonary epithelium
- Ocular epithelial and endothelial barriers
- Placental barrier
- Renal epithelium
- 3D in vitro models
- Microfluidic artificial organs

Methods of evaluation

Distribution of final grade
Class attendance and discussion: 30%

- Reading assignments
- Active class discussion
- Class attendance

Midterm essay/review article: 30%
Final presentation: 40%

Course policies

Academic honesty: As a student of University of Illinois at Chicago (UIC), you have agreed to abide by the University's academic honesty policy, "A Culture of Honesty," and the Student Honor Code. All academic work must meet the standards described in "A Culture of Honesty". Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation. Questions related to course assignments and the academic honesty policy should be directed to the instructor.

Disability services: Concerning disabled students, the University of Illinois at Chicago is committed to maintaining a barrier-free environment so that individuals with disabilities can fully access programs, courses, services, and activities at UIC. Students with disabilities who require accommodations for full access and participation in UIC Programs must be registered with the Disability Resource Center (DRC). Center information can be found at http://www.uic.edu/depts/oar/campus_policies/disability_notification.html

Appropriate classroom behavior: The purpose of this is to ensure the equal right of learning.

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.